

FILTER DESIGN TEST CASE

Multiplexer Module - Band 5 (850 MHz) for 5G NR Sub-6 GHz Front-End Module
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Multiplexer Module filter targeting Band 5 (850 MHz) for 5G NR Sub-6 GHz Front-End Module. The filter is designed on PZT on Si substrate with a target center frequency of 4793.6 MHz and bandwidth of 166.4 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the 5G NR Sub-6 GHz Front-End Module module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Multiplexer Module
- Frequency Band: Band 5 (850 MHz)
- Application: 5G NR Sub-6 GHz Front-End Module
- Substrate: PZT on Si
- Package: QFN (Quad Flat No-Lead)
- Design Tool: Sonnet EM Suite v16
- Design Center: Cedar Rapids Lab

This specification applies to all variants of the Multiplexer Module design targeting Band 5 (850 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	4793.6 MHz
Bandwidth (3 dB):	166.4 MHz
Insertion Loss Target:	≤ 1.6 dB
Return Loss Target:	≥ 12.0 dB
Out-of-Band Rejection:	≥ 25.0 dB
Isolation (Duplexer):	≥ 30.0 dB
Q Factor:	9773
Temperature Coefficient:	-19.8 ppm/°C
Die Size:	1.3 x 1.7 mm
Wafer Size:	8-inch
Number of Resonators:	9
Electrode Material:	Al

Passivation: SiO2

4. TEST PROCEDURE

1. Open Sonnet EM Suite v16 and load the Multiplexer Module template project.
2. Define the substrate stack: PZT on Si with specified layer thicknesses.
3. Set design targets: center freq 4793.6 MHz, BW 166.4 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.6 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (QFN (Quad Flat No-Lead)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, PZT on Si).
10. Perform on-wafer probing using Rohde & Schwarz ZNA67.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 14381 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.6 dB
- Return loss in passband shall be >= 12.0 dB
- Out-of-band rejection at +/- 250 MHz offset >= 25.0 dB
- Group delay variation across passband shall be <= 4 ns
- Temperature drift over -40°C to +85°C shall be <= 11.9 MHz
- Power handling shall be >= +30 dBm at 1 dB compression
- ESD tolerance >= 1000 V (HBM)
- Die yield on qualification lot >= 96%

6. TEST RESULTS

Measurement Date: 2024-09-04
Devices Tested: 31
Insertion Loss (meas): 1.34 dB
Return Loss (meas): 13.8 dB
Rejection (meas): 27.6 dB
Isolation (meas): 35.5 dB
Q Factor (meas): 9773
Group Delay Variation: 2.9 ns
Power Handling: +28 dBm
Die Yield: 96.6%

Result Classification: CONDITIONAL PASS

7. OBSERVATIONS AND FINDINGS

- a) The Multiplexer Module design demonstrated acceptable performance for Band 5 (850 MHz).
- b) Insertion loss met target specification.
- c) Rejection at near-band offset requires additional resonator tuning.
- d) Group delay flatness improved compared to previous revision.
- e) ESD robustness exceeded specification by 2x margin.

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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